Program: **Diploma in Communication and Computer Networking**	
Course Code: **3324**	Course Title: **Operating System Design**
Semester: **3**	Credits: **3**
Course Category: **Programme core course**	
Periods per week: **3 (L:3 T:0 P:0)**	Periods per semester: **45**

Course Objectives:

- Understand the services of an operating system provides to its users and system itself.
- Apply various CPU scheduling algorithms and recognize the classic synchronization problems.
- Compare methods for handling deadlocks and apply various memory management techniques.
- Describe Storage Management

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic knowledge in Computer systems		Introduction to IT systems Lab	1

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	CognitiveLevel
CO1	Understand the basics of Operating System.	10	Understanding
CO2	Analyze the various CPU scheduling algorithms, Concurrency Control (IPC).	15	Analysing
CO3	Learn the concept of Memory Management	10	Understanding
CO4	Understand Storage Management	10	Understanding
	SeriesTest	2	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		2				
CO2	3		2	3			
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Understand the basics of Operating System**		
M1.01	To understand basics of operating systems.	4	Understanding
M1.02	To understand Operating System Types.	6	Understanding
Contents: **Introduction to Operating System:** Operating Systems Overview- Overview and Functions of Operating Systems, Protection and Security, Operating Systems Structures, Services, System Calls. Operating System Types- batch systems - multiprogramming systems - time sharing - multiprocessor systems - real time systems.(Definitions only)			
CO2	**Analyze the various CPU scheduling algorithms, Concurrency Control (IPC).**		
M2.01	Familiarize with Process and Threads	3	Understanding
M2.02	Explanation of Process Scheduling algorithms.	7	Understanding
M2.03	Concept of inter process communication.	4	Understanding
	SeriesTest–I	1	
Contents: **Process and Threads:** Process and Threads - Process concepts, threads. **Process Scheduling Algorithms :**Context switching, CPU and I/O burst cycles - CPU bound and I/O bound processes- difference between preemptive and non-preemptive scheduling- Scheduling algorithms-First Come First Serve(FCFS), Shortest Job First(SJF), Round Robin(RR), Priority Scheduling, Shortest remaining time (SRT). **Concurrency Control (IPC):** Process synchronization, critical- section problem. classic problems of Synchronization, Concepts of Semaphore, Mutex, Monitor, Event Counters.			

CO3	**Learn the concept of Memory Management.**		
M3.01	To understand address space, memory allocation	4	Understanding
M3.02	To understand Memory Allocation,	2	Understanding
M3.03	Explanation of Virtual memory, page replacement algorithms	4	Applying

Contents:
Memory Management: Address binding, logical versus physical address space, Swapping, Contiguous memory allocation memory mapping and protection, memory allocation, fragmentation. Non-contiguous allocation- paging, segmentation. Virtual memory-Demand Paging, page replacement, Allocation of frames, Thrashing, Allocating Kernel Memory

CO4	**Understand Storage Management**		
M4.01	Explanation of File system Interface, file concept, access methods and directory structure.	2	Understanding
M4.02	Study basic concept of file system structure, implementation and allocation methods.	3	Understanding
M4.03	Explanation about various disk scheduling algorithms.	4	Understanding
	SeriesTest–II	1	

Contents:
Storage management: File system Interface — file concept, access methods, directory structure, File Sharing. File system implementation- file system structure & implementation, directory implementation, allocation methods, free space management; Mass storage management - disk structure, disk scheduling- FCFS, SCAN, LOOK, RAID.

Text/Reference:

T/R	BookTitle/Author
T1	Abraham Silberschatz, Peter Gaer Galvin and Greg Gagne,Operating System Concepts , Wiley Publications-Eighth Edition.
T2	Achyut S Godbole, Operating systems, Mc-GRawhill, Third Edition
T3	Anthony T.Velte , Toby J. Velte Robert Elsenpeter, Cloud computing a practical approach TATA McGraw- Hill , 2010
R1	William Stallings,Operating Systems-Internals and Design Principles, PEARSON Publications- **Seventh Edition**
R2	Rohit Khurana, Vikas Publishing,Operating Systems, Vikas Publishing-Second Edition.

R3	Operating Systems, I. Chandra Mohan, PHI, 2013, ISBN – 9788120347267

Online Resources:

Sl.No	WebsiteLink
1	https://nptel.ac.in/courses/106106144 : Introduction to Operating Systems, IIT Madras
2	http://www.tutorialspoint.com/operating_system/
3	http://courses.cs.vt.edu/~csonline/OS/Lessons/index.html